

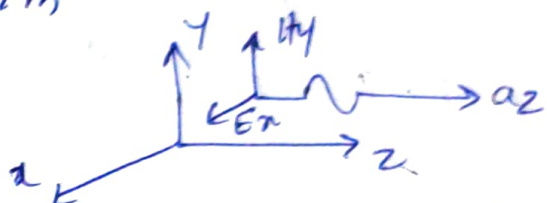
$$D_1 = \epsilon_0 \epsilon_r E_1$$

$$E_1 = \frac{D_1}{\epsilon_0 \epsilon_r}$$

$$= \frac{6.706 \times 10^{-7}}{8.854 \times 10^{-12} \times 4}$$

$$= 18934.94 \text{ V/m}$$

Module - 4



EM waves :- They are synchronized oscillations of electric & magnetic field. The electric & magnetic field of an electromagnetic wave are $\perp$ to each other. They are also $\perp$ to the direction of propagation of EM wave. They travel with a constant velocity of $3 \times 10^8 \text{ m/s}$ in vacuum. They are capable of showing interference. It doesn't require a medium to propagate or travel from one place to another. EM waves can be split into a range of frequencies. This is known as EM spectrum.

Eg: radio waves, microwaves, x rays, gamma rays

Uniform plane waves :- It is the one in which $\vec{E}$ & $\vec{H}$ lie in a plane and they are mutually $\perp$ everywhere in the plane & also have same magnitude everywhere in that plane at any fixed instant.

properties of uniform plane waves in space

- i) At every pt in space $\vec{E}$ & $\vec{H}$ are $\perp$ to each other & $\perp$ to dirⁿ of propagation.
- ii) A uniform plane wave has same value for $\vec{E}$ & $\vec{H}$, everywhere in the plane at any fixed instant. Therefore, for a wave propagating in z -dirⁿ.

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial y} = 0$$

- iii) velocity of propagation in free space $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

$$c = 3 \times 10^8 \text{ m/s}$$

- iv) Ratio of $\vec{E}$ & $\vec{H}$ is constant & is called intrinsic impedance

- v) A uniform EM wave has $\vec{E}$ & $\vec{H}$ components only in dirⁿ $\perp$ to the dirⁿ of propagation. Since $\vec{E}$ & $\vec{H}$ are $\perp$ to the dirⁿ of propagation wave is called transverse wave.

Wave Equation from Maxwell's Equation

EM wave eqn can be explained with Maxwell's eqns.

$$\nabla \times \vec{H} = \vec{J}_c + \frac{\partial \vec{D}}{\partial t} = \alpha \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t} \quad (2)$$

$$\left. \begin{aligned} \vec{D} &= \epsilon \vec{E} \\ \vec{J}_c &= \alpha \vec{E} \end{aligned} \right\}$$

$$\nabla \cdot \vec{B} = \nabla \cdot \vec{H} = 0 \quad (3)$$

$$\nabla \cdot \vec{D} = \nabla \cdot \vec{E} = 0 \quad (4)$$

$$\left\{ \begin{aligned} \nabla \cdot \vec{D} &= \rho_v = 0 \\ \nabla \cdot \vec{E} &= 0 \end{aligned} \right\}$$

Apply curl operation on both sides of equation

$$\nabla \times (\nabla \times \vec{E}) = \nabla \times \left(-\mu \frac{\partial \vec{H}}{\partial t} \right)$$

$$\nabla \times (\nabla \times \vec{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \vec{H}) \quad (5)$$

Subst. (2) in (5)

$$\nabla \times (\nabla \times \vec{E}) = -\mu \frac{\partial}{\partial t} \left(\alpha \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t} \right)$$

$$= -\mu \alpha \frac{\partial \vec{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} \quad (6)$$

By vector identity

$$\nabla \times (\nabla \times \vec{E}) = \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$$

$$\nabla \cdot \vec{E} = 0 \text{ from eqn (4)}$$

$$\nabla \times (\nabla \times \vec{E}) = -\nabla^2 \vec{E}$$

eqn (6) will become

$$-\nabla^2 \vec{E} = -\mu \alpha \frac{\partial \vec{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\boxed{\nabla^2 \vec{E} = \mu \alpha \frac{\partial \vec{E}}{\partial t} + \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}} \quad (7)$$

Now multiply both sides of eqn (7)

$$\nabla^2 (\epsilon \bar{E}) = \mu \alpha \frac{\partial (\epsilon \bar{E})}{\partial t} + \mu \epsilon \frac{\partial^2 (\epsilon \bar{E})}{\partial t^2}$$

$$\boxed{\nabla^2 \vec{D} = \mu \alpha \frac{\partial \vec{D}}{\partial t} + \mu \epsilon \frac{\partial^2 \vec{D}}{\partial t^2}}$$

Equ'n (7) is the 2nd order partial differential equation of $\vec{E}$ and $\vec{E}$ must obey this equation

$\therefore \vec{E}$ is a wave

Apply curl on both sides of equation (2)

$$\begin{aligned} \nabla \times (\nabla \times \vec{H}) &= \nabla \times \left(\alpha \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t} \right) \\ &= \alpha (\nabla \times \vec{E}) + \epsilon \frac{\partial (\nabla \times \vec{E})}{\partial t} \quad \text{--- (8)} \end{aligned}$$

Subst (1) in (8)

$$\begin{aligned} \nabla \times (\nabla \times \vec{H}) &= \alpha \left(-\mu \frac{\partial \vec{H}}{\partial t} \right) + \epsilon \frac{\partial \left(-\mu \frac{\partial \vec{H}}{\partial t} \right)}{\partial t} \\ &= -\mu \alpha \frac{\partial \vec{H}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2} \quad \text{--- (9)} \end{aligned}$$

By vector identity

$$\nabla \times (\nabla \times \vec{H}) = \nabla (\nabla \cdot \vec{H}) - \nabla^2 \vec{H}$$

from equation 3

$$\nabla \cdot \vec{H} = 0$$

$$\nabla \times (\nabla \times \vec{H}) = -\nabla^2 \vec{H}$$

equ'n (9)

$$-\nabla^2 \vec{H} = -\mu \alpha \frac{\partial \vec{H}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

$$\boxed{\nabla^2 \bar{H} = \mu \sigma \frac{\partial \bar{H}}{\partial t} + \mu \epsilon \frac{\partial^2 \bar{H}}{\partial t^2}} \quad \text{--- (10)}$$

mply both sides with μ

$$\nabla^2 (\mu \bar{H}) = \mu \sigma \frac{\partial (\mu \bar{H})}{\partial t} + \mu \epsilon \frac{\partial^2 (\mu \bar{H})}{\partial t^2}$$

$$\boxed{\nabla^2 \bar{B} = \mu \sigma \frac{\partial (\bar{B})}{\partial t} + \mu \epsilon \frac{\partial^2 (\bar{B})}{\partial t^2}}$$

Eqn (10) is the second order partial differential eqn of $\bar{H}$ & $\bar{H}$ must obey the eqn

$\therefore \bar{H}$ is a wave

wave equation in phasor form

Phasor representation of a time harmonic field (varies periodically / sinusoidally with time). Sinusoids can be easily represented as phasors.

$$\nabla^2 \bar{E} = \mu \sigma \frac{\partial \bar{E}}{\partial t} + \mu \epsilon \frac{\partial^2 \bar{E}}{\partial t^2} \quad \text{--- (1)}$$

$$\nabla^2 \bar{H} = \mu \sigma \frac{\partial \bar{H}}{\partial t} + \mu \epsilon \frac{\partial^2 \bar{H}}{\partial t^2} \quad \text{--- (2)}$$

$$\text{Put } \frac{\partial}{\partial t} = j\omega$$

$$\nabla^2 \bar{E} = \mu \sigma j\omega \bar{E} + \mu \epsilon (j\omega)^2 \bar{E}$$

$$= j\mu \omega \sigma \bar{E} + j^2 \mu \epsilon \omega^2 \bar{E}$$

$$= \underbrace{j\mu \omega (\sigma + j \epsilon \omega)}_{\gamma^2} \bar{E}$$

$$\boxed{\nabla^2 \vec{E} = \gamma^2 \vec{E}} \rightarrow \text{Helmholtz eqn}$$

$$\gamma = \sqrt{j\omega\mu(\sigma + j\epsilon\omega)}$$

put $\frac{\partial}{\partial t} = j\omega$ in eqn (2)

$$\begin{aligned}\nabla^2 \vec{H} &= \mu\sigma j\omega \vec{H} + \mu\epsilon(j\omega)^2 \vec{H} \\ &= j\omega\mu\sigma \vec{H} + j^2\mu\epsilon\omega^2 \vec{H} \\ &= j\omega\mu(\sigma + j\epsilon\omega) \vec{H}\end{aligned}$$

$$\boxed{\nabla^2 \vec{H} = \gamma^2 \vec{H}} ; \text{ where } \gamma = \sqrt{j\omega\mu(\sigma + j\epsilon\omega)}$$

not in
syllabus

Circular solution of wave eqn

$$\nabla^2 \vec{E} = \gamma^2 \vec{E}$$

$$\nabla^2 \vec{H} = \gamma^2 \vec{H}$$

Consider $\vec{E}$ is in x direction & $\vec{H}$ is in y direction
wave propagates in z direction.

$$E_y = E_z = 0 \quad (E_x \neq 0)$$

$$H_x = H_z = 0 \quad (H_y \neq 0)$$

In case of wave $\frac{\partial}{\partial x} = \frac{\partial}{\partial y} = 0 \quad (\frac{\partial}{\partial z} \neq 0)$

$$\nabla^2 \vec{E} = \gamma^2 \vec{E}$$

$$\frac{\partial^2}{\partial z^2} E_x = \gamma^2 E_x \quad \text{--- (1)}$$

$$\nabla^2 \vec{H} = -\gamma^2 \vec{H} \quad \text{--- (1)}$$

$$\frac{\partial^2 H_y}{\partial z^2} = -\gamma^2 H_y \quad \text{--- (2)}$$

Solution of eqn (1)

$$\boxed{E_x(z, t) = E_m^+ e^{-\alpha z} \cos(\omega t - \beta z) + E_m^- e^{\alpha z} \cos(\omega t + \beta z)}$$

E_m^+ & E_m^- implies travelling wave in +ve direction
& travelling wave in -ve direction resp.

$$H_y(z, t) = H_m^+ e^{-\alpha z} \cos(\omega t - \beta z - \theta_n) + H_m^- e^{\alpha z} \cos(\omega t + \beta z - \theta_n)$$

θ_n : Phase angle diff. b/w E & H

γ : propagation const. $(\alpha + j\beta)$

β : phase shift const.

α : attenuation const.

(Amplitude of signals get reduced)
Attenuation constant α

It's the attenuation of an EM wave propagating through a medium per unit distance from source. It is the real part of the propagation constant & is measured in nepers/meter. Practically expressed in dB.

$$1 \text{ Np} = 8.686 \text{ dB}$$

Phase shift constant β

When the EM wave propagates phase of E & H rotates. The angular radians rotated per meter travel of the wave is called the phase shift constant. It is the imaginary part of the propagation constant & is measured in radians/meter.

Propagation constant (γ)

EM wave propagate in a sinusoidal fashion. The measure of change in its amplitude & phase per unit distance is called propagation constant.

$$\gamma = \alpha + j\beta$$

Intrinsic impedance η

The ratio of amplitude of E & H of the waves in either direction is called intrinsic impedance of the material in which wave is travelling. Denoted by η .

$$\eta = \frac{E_m^+}{H_m^+} = - \frac{E_m^-}{H_m^-}$$

$$\eta = \sqrt{\frac{\mu}{\epsilon}} \quad \Omega \quad ; \text{ for free space } \eta_0 = 377 \Omega$$

$E_m^+ \rightarrow$ Amplitude of wave component of E travelling

in +ve direction

$E_m^- \rightarrow$ component in -ve direction.

$\Rightarrow$ wave length $\lambda = \frac{2\pi}{\beta} \text{ m}$

$\Rightarrow$ wave velocity

$$V_p = f \lambda \text{ m/s}$$

$$V_g = \frac{\omega}{\beta} \text{ m/s}$$

$$V = \frac{1}{\sqrt{\mu\epsilon}} \text{ m/s}$$

Plane wave propagation in Lossy dielectric medium ($\alpha \neq 0$, $\epsilon = \epsilon_r \epsilon_0$, $\mu = \mu_r \mu_0$)

I. wave equations :-

$$\nabla^2 \bar{E} = \gamma^2 \bar{E}$$

$$\nabla^2 \bar{H} = \gamma^2 \bar{H}$$

Solutions are

$$E_x(z,t) = E_m^+ e^{-\alpha z} \cos(\omega t - \beta z) + E_m^- e^{\alpha z} \cos(\omega t + \beta z)$$

$$H_y(z,t) = H_m^+ e^{-\alpha z} \cos(\omega t - \beta z - \theta_n) + H_m^- e^{\alpha z} \cos(\omega t + \beta z)$$

II. Intrinsic Impedance

Modified Faraday's law

$$\nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t} = -\mu \frac{\partial \bar{H}}{\partial t}$$

$$\begin{vmatrix} a_x^{\wedge} & a_y^{\wedge} & a_z^{\wedge} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = -\mu \frac{\partial}{\partial t} [H_x a_x^{\wedge} + H_y a_y^{\wedge} + H_z a_z^{\wedge}]$$

Assume $\vec{E}$ is in x diren, $\vec{H}$ is in y diren.
Let the direction of propagation be in z diren.

$$E_y = E_z = 0$$

$$H_x = H_z = 0$$

$$\begin{vmatrix} a_x^{\wedge} & a_y^{\wedge} & a_z^{\wedge} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & 0 & 0 \end{vmatrix} = -\mu \frac{\partial}{\partial t} [0 a_x^{\wedge} + H_y a_y^{\wedge} + 0 a_z^{\wedge}]$$

$$-a_y^{\wedge} \left[-\frac{\partial}{\partial z} E_x \right] = -\mu \frac{\partial H_y}{\partial t} a_y^{\wedge}$$

$$\frac{\partial E_x}{\partial z} = -\mu \frac{\partial H_y}{\partial t} \quad (1)$$

Let the wave be in +ve z diren, then

$$E_x = E_m^+ e^{-\gamma z}$$

$$\frac{\partial E_x}{\partial z} = E_m^+ (-\gamma) e^{-\gamma z}$$

$$\frac{\partial E_x}{\partial z} = -\gamma E_m^+ e^{-\gamma z}$$

$$\frac{\partial E_x}{\partial z} = -\gamma E_x \quad (2)$$

Put (2) in (1)

$$-rE_x = -u \frac{\partial Hy}{\partial t}$$

$$rE_x = u \frac{\partial Hy}{\partial t}$$

$$\text{Put } \frac{\partial}{\partial t} = j\omega$$

$$rE_x = u j\omega Hy$$

$$\eta = \frac{E_x}{Hy} = \frac{uj\omega}{r} \quad \left\{ r = \sqrt{j\omega\mu(\alpha + j\omega\epsilon)} \right.$$

Intrinsic impedance $\eta = \frac{E_x}{Hy}$

$$\eta = \frac{j\omega u}{\sqrt{j\omega\mu(\alpha + j\omega\epsilon)}} \quad ; \frac{\alpha}{\omega\epsilon} = \text{tangent loss}$$

$$\eta = \sqrt{\frac{(j\omega\mu)^2}{j\omega\mu(\alpha + j\omega\epsilon)}} \quad \text{Since } \alpha \neq 0, \eta \text{ will become a complex}$$

$$\boxed{\eta = \sqrt{\frac{j\omega\mu}{\alpha + j\omega\epsilon}} = |\eta| \angle \theta_n} \quad \Omega$$

$$\boxed{\theta_n = \frac{1}{2} \left[\frac{\pi}{2} - \tan^{-1} \left(\frac{\omega\epsilon}{\alpha} \right) \right] \text{ rad}}$$

θ_n - phase angle b/w E & H . It depends on properties of medium as well as freq of signal.

$$0 < \theta_n < \pi/4$$

$$(\text{low freq} \rightarrow \theta_n = \pi/4)$$

$$\text{high freq} \rightarrow \theta_n = 0$$

III. Propagation const. γ

$$\gamma = \alpha + j\beta = \sqrt{j\omega\mu(\alpha + j\omega\epsilon)} \quad \text{--- (1)}$$

Squaring both sides

$$\alpha^2 - \beta^2 + 2j\alpha\beta = j\omega\mu(\alpha + j\omega\epsilon)$$

$$\alpha^2 - \beta^2 + 2j\alpha\beta = -\omega^2\epsilon\mu + j\omega\mu\alpha$$

Equating real & imaginary parts

$$\alpha^2 - \beta^2 = -\omega^2\epsilon\mu \quad \text{--- (2)}$$

$$2\alpha\beta = \omega\mu\alpha \quad \text{--- (3)}$$

$$\alpha = \frac{\omega\mu\alpha}{2\beta} \quad \text{--- (4)} \quad \beta = \frac{\omega\mu\alpha}{2\alpha} \quad \text{--- (5)}$$

put (4) in (2)

$$\frac{\omega^2\mu^2\alpha^2}{4\beta^2} - \beta^2 = -\omega^2\epsilon\mu$$

$$\omega^2\mu^2\alpha^2 - 4\beta^4 = -4\omega^2\beta^2\epsilon\mu$$

$$4\beta^4 - 4\omega^2\beta^2\epsilon\mu - \omega^2\mu^2\alpha^2 = 0$$

solving,

$$\beta^2 = \frac{4\omega^2\epsilon\mu \pm \sqrt{16\omega^4\epsilon^2\mu^2 - [4 \times 4 (-\omega^2\mu^2\alpha^2)]}}{2 \times 4}$$

$$\beta^2 = \frac{4\omega^2 \epsilon \mu \pm \sqrt{16\omega^4 \epsilon^2 \mu^2 + 16\omega^2 \mu^2 \alpha^2}}{8}$$

$$= \frac{4\omega^2 \epsilon \mu \pm \sqrt{16\omega^4 \epsilon^2 \mu^2 \left(1 + \frac{\alpha^2}{\omega^2 \epsilon^2}\right)}}{8}$$

$$= \frac{4\omega^2 \epsilon \mu \pm 4\omega^2 \epsilon \mu \sqrt{1 + \left(\frac{\alpha}{\omega \epsilon}\right)^2}}{8}$$

$$\beta^2 = \frac{\omega^2 \epsilon \mu}{2} \pm \frac{\omega^2 \epsilon \mu}{2} \sqrt{1 + \left(\frac{\alpha}{\omega \epsilon}\right)^2}$$

$$= \frac{\omega^2 \epsilon \mu}{2} \left[1 \pm \sqrt{1 + \left(\frac{\alpha}{\omega \epsilon}\right)^2} \right] \quad \text{--- (6)}$$

$$\boxed{\beta = \omega \sqrt{\frac{\epsilon \mu}{2} \left[1 \pm \sqrt{1 + \left(\frac{\alpha}{\omega \epsilon}\right)^2} \right]}}$$

put (6) in (2)

$$\alpha^2 - \frac{\omega^2 \epsilon \mu}{2} \left[1 + \sqrt{1 + \left(\frac{\alpha}{\omega \epsilon}\right)^2} \right] = -\omega^2 \epsilon \mu$$

$$\alpha^2 - \frac{\omega^2 \epsilon \mu}{2} - \frac{\omega^2 \epsilon \mu}{2} \sqrt{1 + \left(\frac{\alpha}{\omega \epsilon}\right)^2} = -\omega^2 \epsilon \mu$$

$$\alpha^2 = -\omega^2 \epsilon \mu + \frac{\omega^2 \epsilon \mu}{2} + \frac{\omega^2 \epsilon \mu}{2} \sqrt{1 + \left(\frac{\alpha}{\omega \epsilon}\right)^2}$$

$$\alpha^2 = -\frac{\omega^2 \epsilon \mu}{2} + \frac{\omega^2 \epsilon \mu}{2} \sqrt{1 + \left(\frac{\alpha}{\omega \epsilon}\right)^2}$$

$$\alpha^2 = \frac{\omega^2 \epsilon \mu}{2} \left[\sqrt{1 + \left(\frac{\alpha}{\omega \epsilon}\right)^2} - 1 \right]$$

$$\alpha = \omega \sqrt{\frac{\epsilon \mu}{2} \left[1 + \left(\frac{\alpha}{\omega \epsilon} \right)^2 - 1 \right]}$$

IV

Phase velocity (propagation velocity) v_p

waves can be in the group & such groups are called wave packets. So the velocity with a wave packet travels is called grp velocity. Velocity with which phase of a wave travels - phase velocity

Consider +ve E wave travelling in +ve direction

$$E_x(z, t) = E_m^+ \cos(\omega t - \beta z)$$

for const. phase $(\omega t - \beta z)$ const.

Diff. w.r.t. t

$$\omega - \beta \frac{dz}{dt} = 0 \quad \frac{dz}{dt} = \frac{\omega}{\beta}$$

$$\text{Phase Velocity } v_p = \frac{dz}{dt} = \frac{\omega}{\beta}$$

$$\beta = \frac{2\pi}{\lambda} \quad ; \quad v_p = \frac{\omega \lambda}{2\pi} = \frac{2\pi f \lambda}{2\pi}$$

$$\boxed{v_p = f \lambda}$$

V. Group velocity (v_g)

velocity with which overall envelope shape of wave amplitude - known as modulation propagates

through space.

$$E_x(z, t) = E_m^+ \cos(\omega t - \beta z)$$

Suppose wave has 2 frequencies but have equal amp
at time

$$\omega_1 = \omega_0 + \Delta\omega \quad \omega_2 = \omega_0 - \Delta\omega$$

β corresponding to ω_1 & ω_2 are

$$\beta_1 = \beta_0 + \Delta\beta \quad \beta_2 = \beta_0 - \Delta\beta$$

grp velocity

$$v_g = \frac{\Delta\omega}{\Delta\beta}$$

Plane wave propagation in free space ($\alpha=0, \epsilon=\epsilon_0, \mu=\mu_0$)

∇ wave eqns are

$$\left. \begin{aligned} \nabla^2 \bar{E} &= \mu\alpha \frac{\partial \bar{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \bar{E}}{\partial t^2} \\ \nabla^2 \bar{H} &= \mu\alpha \frac{\partial \bar{H}}{\partial t} + \mu\epsilon \frac{\partial^2 \bar{H}}{\partial t^2} \end{aligned} \right\} \text{general.}$$

In free space $\alpha=0, \epsilon=\epsilon_0, \mu=\mu_0$

$$\left. \begin{aligned} \nabla^2 \bar{E} &= \mu_0 \epsilon_0 \frac{\partial^2 \bar{E}}{\partial t^2} \\ \nabla^2 \bar{H} &= \mu_0 \epsilon_0 \frac{\partial^2 \bar{H}}{\partial t^2} \end{aligned} \right\} \begin{array}{l} \text{wave eqn} \\ \text{in free space.} \end{array}$$

Put $\frac{\partial}{\partial t} = j\omega$ in above eqn

$$\nabla^2 \bar{E} = \mu_0 \epsilon_0 (j\omega)^2 \bar{E} = -\gamma^2 \bar{E}$$

$$\nabla^2 \vec{H} = \mu_0 \epsilon_0 (\omega)^2 \vec{H} = \gamma^2 \vec{H}$$

$$\gamma^2 = \mu_0 \epsilon_0 (\omega)^2 \quad \gamma = j\omega \sqrt{\mu_0 \epsilon_0} = \alpha + j\beta$$

$$\boxed{\begin{matrix} \alpha = 0 \\ \beta = \omega \sqrt{\mu_0 \epsilon_0} \end{matrix}}$$

II. Intrinsic impedance η

For general medium $\eta = \sqrt{\frac{j\omega\mu}{\alpha + j\omega\epsilon}}$

For free space $\alpha = 0, \mu = \mu_0, \epsilon = \epsilon_0$

$$\eta = \sqrt{\frac{j\omega\mu_0}{j\omega\epsilon_0}} = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

η has only real part; $\theta_n \geq 0$, $\vec{E}$ & $\vec{H}$ are in phase.

III. Phase velocity

$$v_p = \frac{\omega}{\beta} \text{ (general)}$$

Free space $\beta = \omega \sqrt{\mu_0 \epsilon_0}$

$$v_p = \frac{\omega}{\omega \sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c \text{ (velocity of light)}$$

Plane wave propagation in lossless dielectric media / perfect dielectric ($\alpha = 0, \epsilon = \epsilon_0 \epsilon_r, \mu = \mu_0 \mu_r$)

I. wave eqns :-

$$\nabla^2 \vec{E} = \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} + \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{H} = \mu \alpha \frac{\partial \vec{H}}{\partial t} + \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

In lossless media

$$\nabla^2 \vec{E} = \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{H} = \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

} wave eqns
in lossless
dielectric
media

put $\frac{\partial}{\partial t} = j\omega$

$$\nabla^2 \vec{E} = \mu \epsilon (j\omega)^2 \vec{E} = \gamma^2 \vec{E}$$

$$\nabla^2 \vec{H} = \mu \epsilon (j\omega)^2 \vec{H} = \gamma^2 \vec{H}$$

$$\gamma^2 = \mu \epsilon (j\omega)^2 \quad \gamma = j\omega \sqrt{\mu \epsilon}$$

$\gamma = \alpha + j\beta$ comparing $\boxed{\begin{matrix} \alpha = 0 \\ \beta = \omega \sqrt{\mu \epsilon} \end{matrix}}$

I Intrinsic Impedance η

$$\eta = \sqrt{\frac{j\omega\mu}{\alpha + j\omega\epsilon}} \quad (\text{general})$$

For lossless media

$$\eta = \sqrt{\frac{j\omega\mu}{j\omega\epsilon}} = \sqrt{\frac{\mu}{\epsilon}}$$

η has only real part i.e. $\theta_\eta = 0$ ($\vec{E}$ & $\vec{H}$) are in phase

II Phase velocity

$$V_p = \frac{\omega}{\beta} \quad (\text{general medium})$$

$$\beta = \omega \sqrt{\mu \epsilon}$$

$$V_p = \frac{\omega}{\omega \sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\mu \epsilon}}$$

In lossless dielectric $\alpha \ll \omega\epsilon$ is a sp. case
(practical dielectric)

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \quad (\text{general})$$

$$\gamma = \sqrt{j\omega\mu(j\omega\epsilon)(1 + \frac{\sigma}{j\omega\epsilon})}$$

$$\gamma = j\omega\sqrt{\mu\epsilon}(1 + \frac{\sigma}{j\omega\epsilon})^{1/2}$$

Using Binomial Theorem

$$(1+x)^n = 1 + nx \quad \text{neglecting higher order}$$

$$\gamma = j\omega\sqrt{\mu\epsilon} \left(1 + \frac{\sigma}{j\omega\epsilon}\right)^{1/2}$$

$$\gamma = j\omega\sqrt{\mu\epsilon} \left(1 + \frac{1}{2} \cdot \frac{\sigma}{j\omega\epsilon}\right)$$

$$\gamma = j\omega\sqrt{\mu\epsilon} + \frac{j\omega\sqrt{\mu\epsilon}\sigma}{2j\omega\epsilon}$$

$$\gamma = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} + j\omega\sqrt{\mu\epsilon} \quad ; \quad \gamma = \alpha + j\beta$$

$$\alpha = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} \quad \text{nepers/m}$$

$$\beta = \omega\sqrt{\mu\epsilon} \quad \text{rad/m}$$

Plane wave propagation in good conductor

$$(\alpha \approx \omega, \epsilon \approx \epsilon_0, \mu = \mu_r \mu_0 \text{ or } \alpha \gg \omega\epsilon_0, \frac{\sigma}{\omega\epsilon_0} \gg 1)$$

$$\nabla^2 \vec{E} = \gamma^2 \vec{E}$$

$$\gamma = \sqrt{j\mu\omega (\alpha + j\omega\epsilon)} = \alpha + j\beta$$

$$= \sqrt{j\mu\omega\epsilon} \quad \left\{ \sqrt{j} = \sqrt{1\angle 90^\circ} = 1\angle 45^\circ \right\}$$

$$= \sqrt{\mu\omega\epsilon} \sqrt{j}$$

$$= \sqrt{\mu\omega\epsilon} \angle 45^\circ$$

$$= \sqrt{\mu\omega\epsilon} [\cos 45^\circ + j \sin 45^\circ]$$

$$= \sqrt{\mu\omega\epsilon} \left[\frac{1}{\sqrt{2}} + j \frac{1}{\sqrt{2}} \right] = \alpha + j\beta$$

$$\boxed{\alpha = \sqrt{\frac{\mu\omega\epsilon}{2}}}$$

$$\boxed{\beta = \sqrt{\frac{\mu\omega\epsilon}{2}}}$$

$$\mu = \frac{\omega}{\beta} = \omega \sqrt{\frac{2}{\mu\omega\epsilon}}$$

$$\mu = \sqrt{\frac{2\omega^2}{\mu\omega\epsilon}}$$

$$\boxed{\mu = \sqrt{\frac{2\omega}{\epsilon\omega\epsilon}}}$$

intrinsic impedance

$$\eta = \frac{j\mu\omega}{\gamma}$$

$$\eta = \sqrt{\frac{j^2 \mu^2 \omega^2}{\gamma^2}} = \sqrt{\frac{j^2 \mu^2 \omega^2}{j\mu\omega\epsilon}} = \sqrt{\frac{j\mu\omega}{\epsilon}} = \sqrt{\frac{\mu\omega}{\epsilon}} \angle 45^\circ$$

$$\boxed{\eta = \sqrt{\frac{\mu\omega}{\epsilon}} \angle 45^\circ}$$

$$\theta_\eta = 45^\circ$$

Poynting Vector & Poynting Theorem

EM waves carry with them em power. Energy being transported through space during em wave propagation. Relation b/w rate at which energy is transferred & $\vec{E}$ & $\vec{B}$ is given by Poynting Theorem.

As $\vec{E}$ & $\vec{H}$ are vectors to get power density either dot product or 'x' product. But power always flows in a certain direction, vector qty.

$\downarrow$ scalar $\downarrow$ vector.
ex: power radiated from antenna.

$$\text{Power density } \vec{P} = \vec{E} \times \vec{H}$$

$\rightarrow$ Poynting Vector.

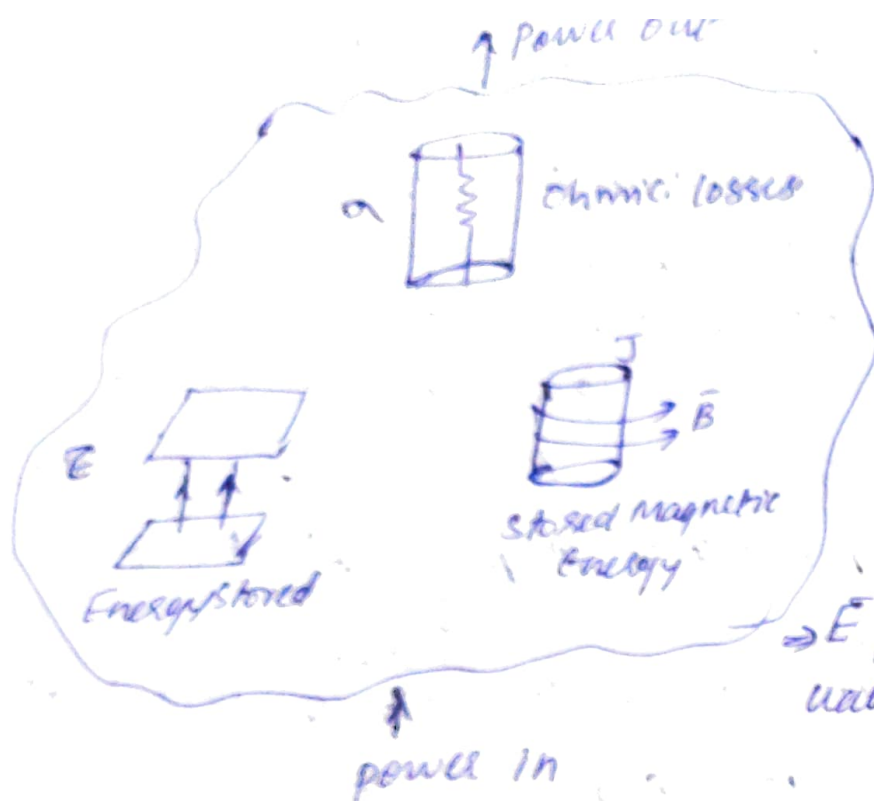
$\vec{P} \rightarrow$ Instantaneous power density vector associated with power flow at the point.

$$\vec{E} \rightarrow \text{V/m} \quad \vec{H} \rightarrow \text{A/m} \quad \vec{E} \times \vec{H} \rightarrow \text{W/m}^2$$

$\Rightarrow$ Poynting theorem ~~stat~~ is based on Law of conservation of energy in electromagnetism.

Poynting theorem states that

'Net power flowing out of a given vol. V is = the ^{time} rate of $\downarrow$ in the energy stored within the volume V minus the ohmic power dissipated



Suppose

$$\vec{E} = E \hat{a}_n$$

$$\vec{H} = H \hat{a}_y$$

$$\vec{P} = \vec{E} \times \vec{H}$$

$$= E \hat{a}_n \times H \hat{a}_y$$

$$= P_z \hat{a}_z$$

$\Rightarrow \vec{E}, \vec{H}$ & $\vec{P}$ are mutually perpendicular.

$\Rightarrow \vec{P}$ is called Poynting vector. To get a net power flowing out any surface, $\vec{P}$ is integrated over the same closed surface.

Integral form of Poynting Theorem

$$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S} = - \frac{\partial}{\partial t} \int_V \left[\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right] dV$$

Total power leaving volume

$$- \frac{\partial}{\partial t} \int_V \epsilon E^2 dV$$

ohmic power dissipated

Rate of decrease in energy stored in EM field

Proof :-

Let's consider Maxwell's eqn based on Ampere's circuital law

$$\nabla \times \vec{H} = \vec{J}_c + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \times \vec{H} = \alpha \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t}$$

Dot the eqn with $\vec{E}$

$$\vec{E} \cdot (\nabla \times \vec{H}) = \vec{E} \cdot (\alpha \vec{E}) + \vec{E} \cdot \left(\epsilon \frac{\partial \vec{E}}{\partial t} \right)$$

$$\left[\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B}) \right]$$

By vector identity

$$\begin{aligned} \vec{E} \cdot (\nabla \times \vec{H}) &= \vec{H} \cdot (\nabla \times \vec{E}) - \nabla \cdot (\vec{E} \times \vec{H}) \\ &= \vec{H} \cdot \left(-\mu \frac{\partial \vec{H}}{\partial t} \right) - \nabla \cdot (\vec{E} \times \vec{H}) \end{aligned}$$

Then

$$\vec{H} \cdot \left(-\mu \frac{\partial \vec{H}}{\partial t} \right) - \nabla \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot (\alpha \vec{E}) + \vec{E} \cdot \left(\epsilon \frac{\partial \vec{E}}{\partial t} \right)$$

$$\nabla \cdot (\vec{E} \times \vec{H}) = -\vec{E} \cdot \left(\epsilon \frac{\partial \vec{E}}{\partial t} \right) - \vec{H} \cdot \left(\mu \frac{\partial \vec{H}}{\partial t} \right) - \vec{E} \cdot (\alpha \vec{E})$$

$$\nabla \cdot (\vec{E} \times \vec{H}) = -\epsilon \frac{\partial E^2}{\partial t} - \frac{\mu}{2} \frac{\partial H^2}{\partial t} - \alpha E^2$$

Taking vol. integral

$$\int_V \nabla \cdot (\vec{E} \times \vec{H}) dV = -\frac{\partial}{\partial t} \int_V \left(\frac{\epsilon}{2} E^2 + \frac{1}{2} \mu H^2 \right) dV - \int_V \alpha E^2 dV$$

$$\left[\frac{\partial}{\partial t} (\vec{H} \cdot \vec{H}) = \vec{H} \cdot \frac{\partial \vec{H}}{\partial t} + \frac{\partial \vec{H}}{\partial t} \cdot \vec{H} = 2 \vec{H} \cdot \frac{\partial \vec{H}}{\partial t} \right]$$

Applying divergence theorem on L.H.S

$$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s} = -\frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dV \\ - \int_V \sigma E^2 dV$$

Hence proof.

Average power density

consider an electric field $\vec{E}$ propagates in free space & is given by

$$\vec{E} = E_m \cos(\omega t - \beta z) \hat{a}_n$$

In a medium, the ratio of magnitudes of $\vec{E}$ & $\vec{H}$ depends on its intrinsic impedance η

$$\eta = \frac{E_m}{H_m}$$

For free space $\eta = \eta_0 = 377 \Omega$

In free space, the magnetic field propagates with speed of light & is given by

$$\vec{H} = H_m \cos(\omega t - \beta z) \hat{a}_y$$

We know $\frac{E_m}{H_m} = \eta_0$

$$\vec{H} = \frac{E_m}{\eta_0} \cos(\omega t - \beta z) \hat{a}_y$$

acc. to Poynting theorem

$$\vec{P} = \vec{E} \times \vec{H}$$

$$\vec{P} = E_m \cos(\omega t - \beta z) \hat{a}_x \times \frac{E_m}{\eta_0} \cos(\omega t - \beta z) \hat{a}_y$$

$$\vec{P} = \frac{E_m^2}{\eta_0} \cos^2(\omega t - \beta z) \hat{a}_z \text{ W/m}^2$$

This is complex Poynting vector

This is power density & power passing through particular area

Power = Power density $\times$ Area (W)

$$\begin{aligned} \text{Pavg} \text{ Pavg} &= \frac{1}{T} \int_0^T \frac{E_m^2}{\eta} \cos^2(\omega t - \beta z) dt \\ &= \frac{E_m^2}{\eta T} \int_0^T \cos^2(\omega t - \beta z) dt \end{aligned}$$

$$\begin{aligned} \text{Pavg} &= \frac{E_m^2}{\eta T} \int_0^T \frac{1 + \cos 2(\omega t - \beta z)}{2} dt \\ &= \frac{E_m^2}{2\eta T} \left[\int_0^T dt + \int_0^T \cos 2(\omega t - \beta z) dt \right] \\ &= \frac{E_m^2}{2\eta T} \left[t \Big|_0^T + \sin \frac{2(\omega t - \beta z)}{2\omega} \Big|_0^T \right] \end{aligned}$$

$$= \frac{E_m^2}{2\eta T} \left[T + \frac{\sin 2(\omega T - \beta z)}{2\omega} + \frac{\sin(2\beta z)}{2\omega} \right]$$

$$\omega = \frac{2\pi}{T} \Rightarrow \omega T = 2\pi$$

$$P_{avg} = \frac{E_m^2}{2\eta T} \left[T + \frac{\sin 2(2\pi - \beta z)}{2\omega} + \frac{\sin(2\beta z)}{2\omega} \right]$$

$$= \frac{E_m^2}{2\eta T} \left[T + \frac{\sin(4\pi - 2\beta z)}{2\omega} + \frac{\sin(2\beta z)}{2\omega} \right]$$

$$= \frac{E_m^2}{2\eta T} \left[T - \frac{\sin 2\beta z}{2\omega} + \frac{\sin 2\beta z}{2\omega} \right]$$

$$= \frac{E_m^2}{2\eta T} \times T = \frac{E_m^2}{2\eta}$$

$$\boxed{P_{avg} = \frac{E_m^2}{2\eta}} \rightarrow \text{Avg. Power density.}$$

Skin depth (skin effect) - (depth of penetration)

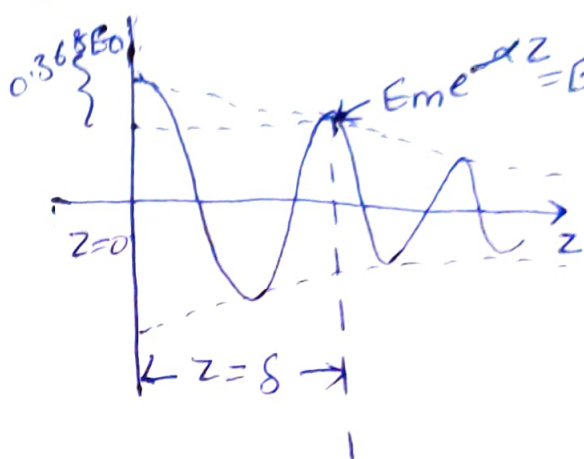
considers the component of electric field E_x travelling in +ve z direction. When it travels in good conductor, the conductivity is high & the attenuation constant α is also high.

We can write such a component in phasor form.

$$E_x = E_m e^{-\alpha z} e^{-j\beta z} \quad - (1)$$

As $\vec{E}$ & $\vec{H}$ travel in a conducting medium its amp. is getting attenuated by a factor of $e^{-\alpha z}$.

Such a wave is shown below.



At $z=0$ amplitude of component E_x is E_m while at $z=s$, amplitude is $E_m e^{-\alpha s}$.

In distance $z=s$, the amplitude gets reduced by factor $e^{-\alpha s}$.

If we select $s = 1/\alpha$, then the factor becomes $e^{-1} = 0.368$. i.e., over a distance $s = 1/\alpha$, the amplitude of the wave decreases approximately 37% of original value.

The distance through which amplitude of the travelling wave decreases to 37% of original amplitude is called skin effect / skin depth / depth of penetration.

$$E_0 e^{-\alpha s} = E_0 e^{-1}$$

$$\alpha s = 1 \quad ; \quad s = 1/\alpha \quad ; \quad s = \frac{1}{\sqrt{\frac{\mu_0 \omega}{2}}} \quad ; \quad \boxed{s = \sqrt{\frac{2}{\mu_0 \omega}}}$$

Skin depth is a measure of the depth to which an EM wave penetrates the medium. The phenomenon by which the field intensity in a conductor rapidly decays is called skin effect.